7.5 Solving Rational Equations

Name _____ Period ____ Date _____

Algebra II 7.5 · Student sheet

I can…

- solve rational equations by cross multiplying.
- solve rational equations using the LCD and check for extraneous solutions.
- use rational equations to solve real-life problems.

Vocabulary · fill in the missing word

Rational equation — an equation that contains rational _____

Cross multiplying — if $\frac{a}{b}$ = _____, then ad = bc

Extraneous solution — a solution of a later step that is not a solution of the _____ equation (it makes a denominator 0)

Part 1 · Paint the room

You can paint a room in 6 hours. A friend can paint it in 3 hours. Guess: working together, how long will it take? _____

Hours (t)	Part you paint	Part your friend paints	Together
1			
2			
3			

Write an equation for working together for t hours and solve it: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Rational equations: One fraction on each side: _____.

Example 1

Solve $\frac{3}{x+1} = \frac{9}{4x+5}$.

More fractions: More fractions: multiply _____ term by the LCD.

Example 2

Solve $\frac{5}{x} + \frac{7}{4} = -\frac{9}{x}$.

Extraneous solutions: An **extraneous solution** makes a _____ equal 0. Always _____ in the original equation.

Example 3

Solve $\frac{x}{x-2} = \frac{2}{x-2} + 3$.

Example 4

Solve $1 - \frac{8}{x-5} = \frac{3}{x}$.

Example 5

A basketball player has made 12 of 30 free throws. How many free throws in a row must she make to raise her free-throw percentage to 60%?

Part 3 · Partner practice

Partner 1

Solve $\frac{4}{x-2} = \frac{6}{x}$.

Partner 2

Find the inverse of $f(x) = \frac{2}{x} - 5$.